Shoaib Asghar

AI/ML Engineer | RAG, LLM Agents & Workflow Automation

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PROFESSIONAL SUMMARY

AI/ML engineer with hands-on experience building Generative AI systems, RAG pipelines, LangGraph AI agents, n8n automation workflows, and computer vision applications. Skilled in LangChain, LangGraph, LLMs, ChromaDB, Supabase/pgvector, OpenCV, and YOLOv8, with a backend foundation in Django, FastAPI, and PostgreSQL. Builds enterprise-grade AI backends that combine vector databases, semantic search, tool-using agents with human-in-the-loop guardrails, and reliable LLM-driven automations.

EXPERIENCE

AI Engineer — DevOrbis **Sep 2026 – Present**

Tech: Python, LangChain, LangGraph, RAG, LLMs, ChromaDB, FastAPI, n8n

- Design, build, and deploy Generative AI and RAG-based solutions, from LLM orchestration to vector search and API delivery.
- Develop and maintain production-ready Python backends that expose ML and LLM capabilities to client-facing applications.

AI/ML Intern — Neuralige **Mar 2026 – Jun 2026**

Tech: LangChain, RAG, LLMs, ChromaDB, OpenRouter, Tavily, DuckDuckGo API, Streamlit, Python, FastAPI, Django

- Designed and built CLARA, an enterprise RAG chatbot using LangChain for LLM orchestration, ChromaDB for vector storage and semantic retrieval, and OpenRouter for multi-model LLM routing. Ranked the #1 intern project at Neuralige.
- Integrated Tavily and DuckDuckGo APIs to augment CLARA with real-time web search, producing a hybrid RAG + live retrieval architecture with dual-source answer synthesis.
- Built REST APIs with FastAPI and Django for production applications, including a JWT-secured Hotel Management System and a Fuel Route Optimizer with geospatial, cost-optimized routing.
- Contributed backend modules to the Mantra AI Therapist app and the U.S.-based RankOne platform in an Agile team (Jira).

PROJECTS

CLARA — Enterprise AI RAG Assistant — *LangChain, RAG, LLMs, ChromaDB, OpenRouter, Tavily*

- Architected a full RAG pipeline (chunking, embeddings, ChromaDB, semantic retrieval, OpenRouter synthesis) with a hybrid layer combining dense retrieval and live web search (Tavily/DuckDuckGo) to reduce hallucination on time-sensitive queries.

Ops Assistant Agent — *LangGraph, LangChain, FastAPI, Supabase, pgvector, React/CLI*

- Built a multi-tool LangGraph agent that answers company questions via a RAG tool over a pgvector knowledge base, looks up structured records (clients, tasks) in Supabase, runs calculations, and drafts emails, with conversation memory across turns.
- Added a human-in-the-loop approval node before any email is sent, plus guardrails (refusal rules, tool-error recovery, max-step limits); authored an annotated Agent Prompt Specification and demoed 3+ multi-step tasks.

CV Screening Pipeline — *n8n, OpenAI/Anthropic, Supabase, Google Drive, Slack, Monday.com*

- Automated an end-to-end hiring workflow: new CVs trigger on Drive/email, text is extracted, and an LLM returns structured screening output (skills, experience, score vs. job description, one-paragraph justification).
- Enforced strict JSON output with a validation branch routing invalid results to manual review; stored results with file links in Supabase, notified Slack/email, created Monday.com items for shortlisted candidates, and added retries with failure alerts. Tested on 5 CVs including a malformed file.

Smart Driver Assistant App (Final Year Project) — *Python, OpenCV, YOLOv8, Raspberry Pi*

- Built a real-time AI driver assistance system using YOLOv8 object detection and OpenCV lane tracking on Raspberry Pi, targeting sub-100ms inference latency for safety-critical alerts.
- Selected for funding by the Head of the CS Department through a UNICEF-supported funding initiative.

Fuel Route Optimizer — *Python, Django REST Framework, OpenRouteService, Geopy, Pandas*

- Built a production-grade API that geocodes multi-stop itineraries, queries OpenRouteService for optimal routes, and computes cost savings, with caching middleware and API rate-limit handling.

Facial Expression Recognition Network — *Python, NumPy, Pandas, Neural Networks*

- Built a multi-class facial emotion classifier from scratch in NumPy (forward/backpropagation, gradient descent, activations), achieving 85%+ test accuracy.

TECHNICAL SKILLS

AI & Generative AI: LangChain, LangGraph, RAG, LLMs, AI Agents, Agentic Workflows, Human-in-the-Loop, Guardrails & Tool-Error Recovery, Prompt Engineering, Structured JSON Output & Schema Validation, Semantic Search, OpenAI/Anthropic APIs, OpenRouter
Vector & Search: ChromaDB, Supabase pgvector, Vector Embeddings, Hybrid Retrieval, Tavily, DuckDuckGo API
Workflow Automation: n8n, Zapier, Webhooks & API Integrations, Slack, Monday.com, Google Drive, Retry/Error Handling & Failure Alerting
Computer Vision & ML: OpenCV, YOLOv8, Neural Networks, Deep Learning, Pandas, NumPy
Backend & Databases: Django, DRF, FastAPI, Flask, Node.js, REST APIs, JWT, PostgreSQL, Supabase, MongoDB, SQL
Languages: Python, SQL, JavaScript, C/C++, Bash
Tools: Git, GitHub, Docker, Postman, Jira, Linux, Streamlit, React, Raspberry Pi

EDUCATION & CERTIFICATION

B.S. Computer Science — Information Technology University (ITU) Punjab **Aug 2022 – Jun 2026**

Coursework: Artificial Intelligence, Machine Learning, Computer Vision, Data Structures & Algorithms, Database Systems, Operating Systems
Certification: Web Development Certificate, Udemy